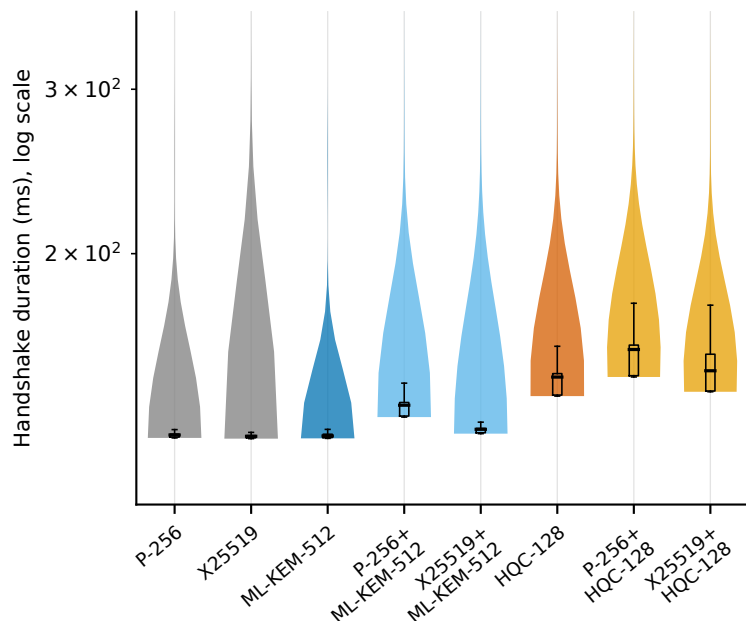
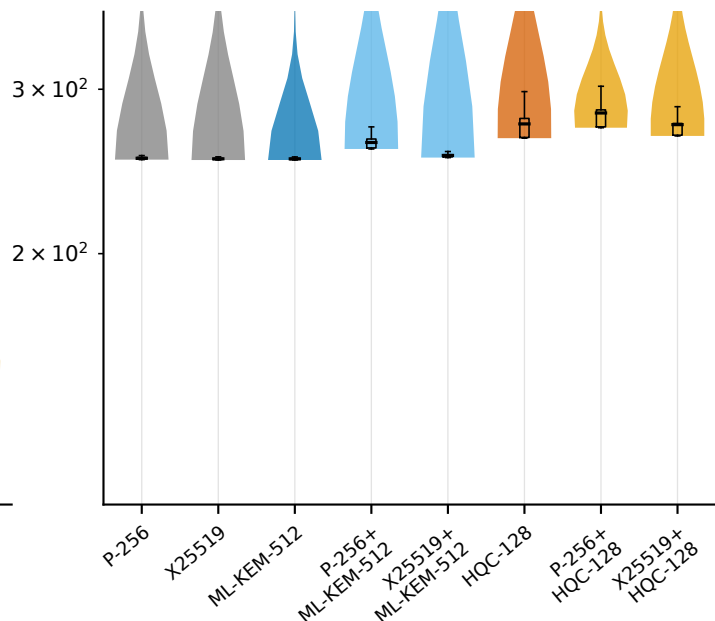


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	138.8	72.1	1.13k	272.8	57.9	2.11k
X25519	150.4	121.9	3.13k	275.3	65.3	2.10k
ML-KEM-512	133.3	55.1	1.28k	264.4	45.4	2.11k
P-256+ML-KEM-512	152.8	87.4	1.66k	294.5	72.2	2.04k
X25519+ML-KEM-512	148.9	97.2	1.28k	290.8	89.1	3.63k
HQC-128	161.4	76.8	1.41k	313.6	67.3	2.23k
P-256+HQC-128	172.6	75.3	1.26k	294.8	36.4	1.43k
X25519+HQC-128	162.3	68.2	1.26k	300.6	54.3	1.42k